

# Power Markets Casebook

v0.24.3

Applied decision cases for turning market, grid, contract, and portfolio concepts into professional screens and memos.

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v0.24.3

The Casebook is where concepts become decisions. Each case gives enough facts to build a professional artifact, then asks for the recommendation, the binding constraint, and the assumption that would change the answer.

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## How To Read

Pick a case when the corresponding lecture idea feels familiar enough to use. If the answer feels hand-wavy, return to the linked lectures and rebuild the artifact.

1. Read the decision question before the scenario data.
2. Build the requested artifact before reading the worked solution.
3. Compare your recommendation to the assumption flip.

## Casebook Structure

The cases move from commercial screens to operating-model choices. Each case is standalone, and together they test whether the concepts can produce usable work.

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### Casebook

## Applied Decision Practice

The cases turn the lectures into work products: a screen, a memo, a scorecard, a product test, or an operating-model choice.

- Use the given facts before adding judgment.
- Find the binding constraint and the first assumption that changes the answer.
- Leave with a reusable artifact, not a paragraph of opinion.

**Start Case 1**

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**Glossary**

## Glossary

An alphabetical reference for the terms, acronyms, market roles, and technical objects used throughout the notes.

- Use it for quick recall.
- Return to the relevant lecture when the term needs mechanism, economics, or context.

[Open Glossary](#)

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## References

## References

The bibliography behind the notes: books, reports, official pages, and named practitioner analyses.

- Use it to trace factual claims back to source families.
- Treat dated market claims as dated; refresh live numbers before operational use.

**Open References**

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## Applied Cases

## Case 1. Shaped Solar-Storage Power Purchase Agreement Quote

References: [1] [2] [3] [4] [5] [6]

A buyer does not ask for “solar” in the abstract. It asks for price certainty, evidence of clean supply, a volume shape, and someone to carry the mismatch between production and consumption. The seller can accept that mismatch, sell a simpler product, or price the shape explicitly. The work product is a quote stack and a [decision memo](#).

### Decision Question

Should a renewable independent power producer (IPP) offer a shaped solar-plus-battery energy storage system (BESS) power purchase agreement (PPA), a simpler [as-produced PPA](#), or no-bid?

### Situation

A corporate buyer wants a 10-year clean-power product for a stable industrial load. The buyer prefers a flat hourly shape because its procurement team wants budget certainty and less exposure to evening prices.

The seller owns a solar project and can add a co-located battery. The commercial team asks for a quote that is attractive to the buyer, acceptable to investment committee, and explicit about who owns residual risk.

### Scenario Data

These are scenario numbers for the case, not market quotes.

| Item                            | Value                                                                         |
|---------------------------------|-------------------------------------------------------------------------------|
| Buyer annual demand             | 100 GWh/year                                                                  |
| Product requested               | 100 gigawatt-hours (GWh)/year flat profile, about 11.4 megawatts (MW) average |
| Solar project output            | 90 GWh/year                                                                   |
| Co-located <a href="#">BESS</a> | 25 MW / 50 megawatt-hours (MWh)                                               |



| Item                                               | Value       |
|----------------------------------------------------|-------------|
| Battery <u>round-trip efficiency</u>               | 88%         |
| As-produced solar offer floor                      | 45 EUR/MWh  |
| Buyer maximum acceptable shaped price              | 68 EUR/MWh  |
| Stress firming premium in hard hours               | 120 EUR/MWh |
| Residual hard-hour volume after solar+ <u>BESS</u> | 20 MWh/h    |
| Number of hard hours in stress case                | 250 h/year  |

For the base quote, use these illustrative quote-stack inputs. They are scenario assumptions for the exercise, not market benchmarks.

| Quote component                              | EUR/MWh |
|----------------------------------------------|---------|
| Raw as-produced clean energy                 | 45.0    |
| Shape and <u>firming cost</u>                | 9.0     |
| Expected <u>imbalance</u> and forecast error | 2.0     |
| Grid, tariff, and settlement administration  | 1.5     |
| Certificates and evidence handling           | 1.0     |
| Credit and tenor allowance                   | 1.5     |
| Battery <u>opportunity cost</u>              | 3.0     |
| Risk margin                                  | 2.0     |

## Product Boundary To Price

Before quoting a number, the seller needs to define the product boundary. A shaped [PPA](#) bundles clean MWh with settlement, evidence, and fallback obligations.

| Term                                  | Case Boundary                                                                                         | Why It Matters                                                                      |
|---------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Quantity                              | about 11.4 MW flat hourly profile, 100 GWh/year commercial target                                     | fixes the shape being sold                                                          |
| Physical source                       | seller solar output plus co-located <a href="#">BESS</a> where available                              | separates asset-backed delivery from market firming                                 |
| Settlement                            | fixed shaped price against the agreed hourly profile                                                  | determines who pays when production and load differ                                 |
| Residual firming                      | seller covers expected residuals; hard-hour tail must be capped, indexed, or passed through           | prevents an uncapped short-volatility position                                      |
| <a href="#">Imbalance</a>             | balance responsible party ( <a href="#">BRP</a> )-managed, with agreed data and performance reporting | connects contract economics to operational evidence                                 |
| Certificates and matching evidence    | defined certificate vintage, geography, delivery period, and data package                             | avoids selling a clean-energy claim without proof rules                             |
| Dispatch rights                       | seller/optimizer retains rights needed to deliver the shape and manage the <a href="#">BESS</a>       | stops the same flexibility being promised elsewhere                                 |
| Credit and <a href="#">collateral</a> | buyer <a href="#">credit support</a> and seller exposure limit stated in the term sheet               | prevents tenor and <a href="#">collateral</a> from being hidden in the energy price |

## Tasks

1. Build a quote stack for the shaped product.
2. Identify which risks remain with the seller.
3. Decide whether to offer the shaped product, offer only as-produced energy, or no-bid.
4. State the first assumption that would change the decision.

## Worked Solution

The quote stack must start with the energy the asset can naturally provide, then add the cost of reshaping that energy into the buyer's requested profile. The quote is usable only if the term sheet assigns the residual mismatch to a party, a price, a cap, or a reopener.

| Component                                    | EUR/MWh     | Risk Owner If Included In Fixed Price |
|----------------------------------------------|-------------|---------------------------------------|
| Raw as-produced clean energy                 | 45.0        | Seller                                |
| Shape and <u>firming cost</u>                | 9.0         | Seller                                |
| Expected <u>imbalance</u> and forecast error | 2.0         | Seller / <u>BRP contract</u>          |
| Grid, tariff, and settlement administration  | 1.5         | Seller                                |
| Certificates and evidence handling           | 1.0         | Seller                                |
| Credit and tenor allowance                   | 1.5         | Seller                                |
| Battery <u>opportunity cost</u>              | 3.0         | Seller                                |
| Risk margin                                  | 2.0         | Seller                                |
| <b>Indicative shaped quote</b>               | <b>65.0</b> |                                       |

At 65 EUR/MWh, the product sits below the buyer's stated 68 EUR/MWh ceiling. That alone is not enough. The tail risk matters.

## Numerical Intuition

The stress firming cost is:

$$20 \text{ MWh/h} \times 250 \text{ h/year} \times 120 \text{ EUR/MWh} = 600,000 \text{ EUR/year}$$

The fixed risk margin in the quote is:

$$100,000 \text{ MWh/year} \times 2 \text{ EUR/MWh} = 200,000 \text{ EUR/year}$$

The quoted margin falls short of the illustrated stress case. The product can still be commercial if the contract caps the seller's hard-hour procurement risk at a fixed price.

## Decision Memo

Recommendation: offer the shaped product at 65 EUR/MWh only with a residual-firming collar.

The seller should own expected shaping and imbalance risk inside the quoted stack. The buyer should accept either a pass-through, cap, index, or renegotiation trigger for residual hard-hour firming beyond the agreed collar. Without that boundary, the product turns into a short volatility position hidden inside a PPA.

The investment-committee test is whether the price, cap, credit terms, evidence package, and dispatch rights describe the same product. If the buyer wants the seller to carry unlimited residual firming, the quote must move up or the product must narrow.

The as-produced fallback should remain on the table at 45 EUR/MWh. It is less valuable to the buyer, but it avoids making a fixed-price promise that the asset cannot physically support under stress.

## Assumption Flip

The recommendation changes if the residual hard-hour volume rises from 20 MWh/h to 35 MWh/h.

$$35 \text{ MWh/h} \times 250 \text{ h/year} \times 120 \text{ EUR/MWh} = 1,050,000 \text{ EUR/year}$$

At that point the seller either needs a much higher fixed price, a portfolio hedge, a third-party firming partner, or a no-bid decision.

## Failure Mode To Avoid

Approve the fixed shaped quote because the expected quote stack is below the buyer's ceiling, then leave hard-hour residuals to be absorbed by the portfolio without a cap, index, or reopener.

That decision leaves the portfolio carrying an unbounded residual exposure. The product promises shape, timing, settlement, and fallback procurement alongside annual clean energy.

## Links Back To The Lectures

- [Load Is Not Energy](#)
  - [Solar Shape](#)
  - [PPAs As Risk-Transfer Machines](#)
  - [PPA Structure Catalog](#)
  - [PPA Pricing](#)
  - [BESS Revenue Stack](#)
  - [Dispatch Optimization](#)
  - [Project Valuation](#)
  - [Future Of PPAs](#)
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# Case 2. The 1h, 2h, Or 4h Battery Decision

References: [7] [8] [9] [10] [11] [5]

Battery duration is an investment claim: the next MWh of energy capacity must earn more than it costs after efficiency, degradation, tariffs, grid rights, and contract obligations. This case asks the duration question as an investment screen rather than as a technology preference.

## Decision Question

For a constrained solar site, should the project build a 1-hour, 2-hour, or 4-hour battery?

## Situation

A solar project has secured land and a grid position. The grid connection can export the solar plant, but adding storage requires choosing a battery duration before the PPA and optimizer contracts are finalized.

The commercial team wants a simple answer. The correct answer is conditional: the first hour, second hour, and later hours do not face the same scarcity distribution.

## Scenario Data

These are scenario assumptions for the case.

| Item                            | Value                         |
|---------------------------------|-------------------------------|
| Solar project size              | 80 MWp                        |
| Grid export limit               | 60 MW                         |
| Battery power rating            | 30 MW                         |
| Candidate durations             | 1h, 2h, 4h                    |
| Carrying charge used for screen | 10% of incremental capex/year |
| <u>Round-trip efficiency</u>    | 88%                           |

| Item                         | Value                                                                                                                                                   |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <u>Degradation allowance</u> | included in net margin                                                                                                                                  |
| Insurance policy             | scenario assumes the modeled operating plan is insurable; if a policy caps cycles or duty cycle below the modeled plan, rerun the screen under that cap |

| Incremental Block      | Incremental Energy | Incremental Capex | Annual Hurdle         |
|------------------------|--------------------|-------------------|-----------------------|
| First hour             | 30 MWh             | 5.0 million EUR   | 0.50 million EUR/year |
| Second hour            | 30 MWh             | 3.5 million EUR   | 0.35 million EUR/year |
| Third and fourth hours | 60 MWh             | 7.0 million EUR   | 0.70 million EUR/year |

| Incremental Block      | Usable Cycles/Year | Net Value Per MWh Moved | Extra <u>Curtailment</u> /Contract Value |
|------------------------|--------------------|-------------------------|------------------------------------------|
| First hour             | 310                | 55 EUR/MWh              | 0.05 million EUR/year                    |
| Second hour            | 240                | 45 EUR/MWh              | 0.15 million EUR/year                    |
| Third and fourth hours | 120                | 25 EUR/MWh              | 0.07 million EUR/year                    |

## Duration Gates

The duration decision needs more than an arbitrage calculation. Each incremental hour must pass the commercial, technical, and contractual gates that make the model value collectible.

| Gate           | What To Check                                         | Failure Signal                                              |
|----------------|-------------------------------------------------------|-------------------------------------------------------------|
| Marginal value | the next MWh of energy capacity clears its own hurdle | average spread from the first hours is applied to all hours |

| Gate                                  | What To Check                                                                                            | Failure Signal                                                                      |
|---------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Cycle, warranty, and insurance budget | modeled throughput is consistent with <u>degradation</u> allowance, warranty terms, and insurance policy | optimizer earns value by consuming life or coverage headroom not priced in the case |
| Grid import and export rights         | <u>connection agreement</u> permits the charging and discharging pattern                                 | battery cannot access the hours that justify the duration                           |
| Product eligibility                   | extra duration qualifies for the intended reserve, firming, or <u>PPA</u> product                        | extra MWh exists physically but earns no contracted premium                         |
| Control rights                        | dispatch rights remain with the party whose economics justify the investment                             | <u>PPA</u> , optimizer, and reserve obligations all assume priority                 |
| <u>Augmentation</u> option            | site layout, transformers, inverters, and permits leave room for later expansion                         | management must choose all duration now under weak evidence                         |

## Tasks

1. Calculate the annual value of each duration block.
2. Compare each block with its annual hurdle.
3. Recommend 1h, 2h, or 4h.
4. State what evidence would change the decision.

## Worked Solution

The first hour value:

$$30 \text{ MWh} \times 310 \text{ cycles/year} \times 55 \text{ EUR/MWh} + 50,000 \text{ EUR/year} \\ = 561,500 \text{ EUR/year}$$



The first hour clears its 500,000 EUR/year hurdle by about 61,500 EUR/year.

The second hour value:

$$\begin{aligned} & 30 \text{ MWh} \times 240 \text{ cycles/year} \times 45 \text{ EUR/MWh} + 150,000 \text{ EUR/year} \\ & = 474,000 \text{ EUR/year} \end{aligned}$$

The second hour clears its 350,000 EUR/year hurdle by about 124,000 EUR/year.

The third and fourth hour value:

$$\begin{aligned} & 60 \text{ MWh} \times 120 \text{ cycles/year} \times 25 \text{ EUR/MWh} + 70,000 \text{ EUR/year} \\ & = 250,000 \text{ EUR/year} \end{aligned}$$

The third and fourth hours miss their 700,000 EUR/year hurdle by about 450,000 EUR/year.

### Numerical Intuition

The best duration is the last block whose marginal value exceeds its marginal cost.

| Option | Annual Value          | Annual Hurdle         | Margin To Hurdle       |
|--------|-----------------------|-----------------------|------------------------|
| 1h     | 0.56 million EUR/year | 0.50 million EUR/year | +0.06 million EUR/year |
| 2h     | 1.04 million EUR/year | 0.85 million EUR/year | +0.19 million EUR/year |
| 4h     | 1.29 million EUR/year | 1.55 million EUR/year | -0.26 million EUR/year |

The 4-hour option creates more gross value than the 2-hour option, but it destroys value at the margin in this scenario.

## Decision Memo

Recommendation: build the 2-hour battery, preserve design optionality for later [augmentation](#), and do not commit to 4 hours until the site has stronger evidence of longer-duration scarcity or contracted value.

The binding constraint is the weak marginal value of hours three and four. The 2-hour asset captures the strongest short-duration price and [curtailment](#) value while avoiding a large capital block with too few high-margin cycles.

The investment decision should approve the 2-hour build and require the design team to preserve practical [augmentation](#) options where the cost is modest: space, cable routes, transformer headroom, inverter/control compatibility, and permitting language. That keeps the project from paying today for duration that is not yet supported by the evidence.

The commercial team should ask for four pieces of evidence before revisiting 4 hours:

- a contract that pays for longer sustained delivery
- tariff or grid rules that reward longer import/export management
- historical or forecast price data showing enough multi-hour scarcity events after [degradation](#) and fees
- warranty, insurance, and [augmentation](#) terms showing that the value can be captured without consuming battery life or coverage headroom faster than priced

## Assumption Flip

The 4-hour decision changes if the third and fourth hours can earn 70 EUR/MWh net value for 160 cycles/year.

$$\begin{aligned} &60 \text{ MWh} \times 160 \text{ cycles/year} \times 70 \text{ EUR/MWh} + 70,000 \text{ EUR/year} \\ &= 742,000 \text{ EUR/year} \end{aligned}$$

It can also change in the other direction if an insurance policy caps eligible operation below the modeled cycle need. In that case, replace the modeled useful cycles with the policy-compatible cycle count before comparing duration blocks. The project may still proceed, but only after the owner obtains different coverage, changes the operating plan, or prices the lost optionality.

That clears the 700,000 EUR/year hurdle. The case for 4 hours is therefore not philosophical. It depends on the frequency and value of longer scarcity windows.

## Failure Mode To Avoid

Choose four hours because the incremental capex per MWh is lower than the first two hours, then apply the average short-duration spread to all four hours.

That decision uses the economics of the valuable first hours to justify weaker later hours. The third and fourth hours need their own scarcity evidence, contract value, tariff benefit, or portfolio hedge value.

## Links Back To The Lectures

- [Frequency And Reserves](#)
  - [Batteries As Time-Shifting Assets](#)
  - [BESS Revenue Stack](#)
  - [Duration Choice](#)
  - [Degradation, Warranties, Battery Life](#)
  - [Long-Duration Versus Short-Duration Storage](#)
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# Case 3. Grid-Constrained Site Screen

References: [11] [12] [13] [14] [15] [16] [17]

Solar development often begins with land and irradiation. Project value often ends with grid access, tariffs, congestion, curtailment, and what the [connection agreement](#) actually permits. The work product is a grid-aware investment screen for three attractive sites.

## Decision Question

Which site should the [IPP](#) prioritize when irradiation, grid rights, tariffs, and portfolio fit point in different directions?

## Situation

The development team has three candidate sites for a new solar-plus-storage project. All three can be developed, but only one can receive near-term investment resources.

The highest-yield site is not automatically the best site. The decision must compare physical production with grid economics and commercial optionality.

## Scenario Data

These are scenario assumptions.

| Metric                               | Site A             | Site B             | Site C             |
|--------------------------------------|--------------------|--------------------|--------------------|
| Solar size                           | 100 MWp            | 90 MWp             | 110 MWp            |
| Expected PV yield                    | 1,060 kWh/kWp/year | 1,000 kWh/kWp/year | 1,030 kWh/kWp/year |
| Earliest firm connection             | 2030               | 2028               | 2028               |
| Export limit                         | 65 MW              | 80 MW              | 55 MW              |
| Expected <a href="#">curtailment</a> | 5%                 | 1%                 | 8%                 |

| Metric                | Site A      | Site B    | Site C     |
|-----------------------|-------------|-----------|------------|
| Grid/tariff/loss cost | 8 EUR/MWh   | 4 EUR/MWh | 11 EUR/MWh |
| BESS import allowed   | conditional | yes       | no         |
| Portfolio zone fit    | weak        | strong    | medium     |
| Permitting complexity | medium      | medium    | low        |

Assume the merchant value before grid costs is 48 EUR/MWh for all sites. Ignore capex differences for the first screen.

## Tasks

1. Calculate net annual MWh after curtailment.
2. Calculate net value per MWh after grid/tariff/loss cost.
3. Estimate annual gross margin before capex and opex.
4. Add a grid-rights and timing screen.
5. Recommend the priority site.

## Worked Solution

Annual MWh before curtailment:

| Site | Calculation                 | Gross Output     |
|------|-----------------------------|------------------|
| A    | 100,000 kWp x 1,060 kWh/kWp | 106,000 MWh/year |
| B    | 90,000 kWp x 1,000 kWh/kWp  | 90,000 MWh/year  |
| C    | 110,000 kWp x 1,030 kWh/kWp | 113,300 MWh/year |

After curtailment:

| Site | Net MWh/year |
|------|--------------|
| A    | 100,700      |
| B    | 89,100       |
| C    | 104,236      |

Net value per MWh:

| Site | Value Before Grid Cost | Grid/Tariff/Loss Cost | Net Value  |
|------|------------------------|-----------------------|------------|
| A    | 48 EUR/MWh             | 8 EUR/MWh             | 40 EUR/MWh |
| B    | 48 EUR/MWh             | 4 EUR/MWh             | 44 EUR/MWh |
| C    | 48 EUR/MWh             | 11 EUR/MWh            | 37 EUR/MWh |

Annual gross margin before capex and opex:

| Site | Net MWh/year | Net Value  | Annual Gross Margin   |
|------|--------------|------------|-----------------------|
| A    | 100,700      | 40 EUR/MWh | 4.03 million EUR/year |
| B    | 89,100       | 44 EUR/MWh | 3.92 million EUR/year |
| C    | 104,236      | 37 EUR/MWh | 3.86 million EUR/year |

## Grid Rights Screen

| Site | Grid And Commercial Signal                                                                     | Investment Meaning                                              |
|------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| A    | highest base margin, but<br>2030 firm connection,<br>conditional BESS import, weak<br>zone fit | value is delayed and the<br>storage strategy is not yet<br>firm |

| Site | Grid And Commercial Signal                                                                    | Investment Meaning                                              |
|------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| B    | lower output, but 2028 connection, low <u>curtailment</u> , full BESS import, strong zone fit | lower headline MWh but better executable optionality            |
| C    | high output, no BESS import, high <u>curtailment</u> and grid cost                            | looks large on MWp but weak for shaped products and flexibility |

The tariff and connection question includes EUR/MWh, reserved capacity, geography, limited access, and producer payments. Nordic tariff work increasingly separates these items. A site screen should therefore ask what capacity is being reserved, when it is firm, whether limited or interruptible access is acceptable, and whether import rights let the battery charge when the commercial case requires it.

### Numerical Intuition

The top-line spread between sites is small after grid effects.

Site A gross margin – Site B gross margin = about 0.11 million EUR/year

That is not enough to dominate a two-year connection delay, weaker portfolio fit, conditional battery import rights, and higher grid uncertainty.

The undiscounted value delayed by choosing Site A over a near-term 2028 site is roughly:

2 years x 4.03 million EUR/year = 8.06 million EUR of gross margin timi

Use this as a screen, rather than a project net present value (NPV), to prevent a small annual yield advantage from hiding a large timing and optionality penalty.

## Decision Memo

Recommendation: prioritize Site B.

Site A has the highest simple annual gross margin, but Site B has the best decision quality. It connects earlier, has lower [curtailment](#), lower grid cost, stronger portfolio fit, and full BESS import rights. Those features make it more useful for shaped PPAs, reserve participation, and portfolio balancing.

The first diligence request should be the grid-rights evidence pack: firm connection date, export capacity, import capacity, curtailment rules, capacity-tariff exposure, producer-payment geography, limited-access terms, and whether BESS operation is allowed under the same [connection agreement](#) or needs a separate approval.

Site A should remain in development if the connection timing improves or if a buyer values the location enough to pay for the grid uncertainty. Site C is not a priority unless its low permitting complexity creates a materially faster path than assumed.

## Assumption Flip

The recommendation changes if Site A obtains firm 2028 connection and full [BESS](#) import rights.

Then the main penalty is grid cost rather than timing and optionality. Site A's higher output may justify prioritization, especially if the portfolio already has enough exposure in Site B's zone.

## Failure Mode To Avoid

Rank the sites by base-case P50 generation value, then treat connection timing, import rights, [curtailment](#), and tariff exposure as diligence items.

That decision puts the grid outside the economics even though the grid determines when the MWh is deliverable, what the battery can do, and which commercial products remain feasible.

## Links Back To The Lectures

- [Wires, Zones, Tariffs](#)



- Voltage, Reactive Power, Inertia
  - Nordic First Pass
  - Grid Connection And Tariff Economics
  - Solar Project Economics
  - Portfolio Modeling And Hedging
  - Grid Bottlenecks As Strategy
-

# Case 4. BESS Revenue Stack Under Saturation

References: [18] [19] [20] [21] [7] [9] [5] [22]

A battery revenue stack is a set of coupled obligations on the same MW, MWh, state of charge, grid rights, cycle budget, insurance policy, and control system. When a service saturates, test which obligations remain valuable after opportunity cost and downside are counted.

## Decision Question

Should the asset owner contract part of the battery’s flexibility, keep it merchant, or reserve optionality while the revenue stack evolves?

## Situation

A 50 MW / 100 MWh battery can participate in energy arbitrage and reserve products. A utility offers a five-year floor contract for part of the asset. An optimizer offers full merchant dispatch with no guaranteed payment. A customer team asks whether part of the battery should be reserved for shaped PPA delivery.

The project cannot sell the same flexibility three times.

## Scenario Data

These are scenario assumptions.

| Item                          | Value                 |
|-------------------------------|-----------------------|
| Battery                       | 50 MW / 100 MWh       |
| <u>Round-trip efficiency</u>  | 88%                   |
| Expected merchant stack value | 1.70 million EUR/year |
| Downside merchant stack value | 0.70 million EUR/year |
| Floor contract fixed payment  | 0.90 million EUR/year |

| Item                                                    | Value                                                                                                         |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Expected upside retained under floor                    | 0.30 million EUR/year                                                                                         |
| Shaped <u>PPA</u> flexibility premium                   | 0.75 million EUR/year                                                                                         |
| Merchant value left after reserving <u>PPA</u> capacity | 0.80 million EUR/year                                                                                         |
| Lender-counted share of pure merchant value             | 20% scenario assumption                                                                                       |
| Lender-counted share of fixed floor                     | 100% scenario assumption                                                                                      |
| Insurance cycle limit                                   | scenario assumes no binding limit beyond the modeled cycle budget; verify policy terms before using the stack |

## Tasks

1. Compare expected value and financeable value.
2. Identify which option has the strongest downside protection.
3. State the opportunity cost of reserving capacity for a shaped PPA.
4. Recommend an allocation.

## Contracting Routes

Recent BESS offtake discussions use several structures rather than one generic “battery revenue” line. The case reduces them to four routes the investment committee can compare.

| Route                      | What Is Sold                                                          | Control-Rights Consequence                                      | When It Fits                                     |
|----------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------|
| Full merchant optimization | optimizer pursues energy, reserve, and <u>imbalance</u> opportunities | asset owner keeps upside and downside; lender credit is limited | owner accepts volatility and has strong controls |

| Route                               | What Is Sold                                                                                                | Control-Rights Consequence                                          | When It Fits                                                                         |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Floor or revenue-share contract     | floor, toll, revenue swap, or minimum guaranteed payment                                                    | some upside and dispatch discretion are transferred                 | project needs financeability or downside protection                                  |
| Partial hybrid <a href="#">PPA</a>  | defined battery flexibility supports a customer profile, residual remains merchant or contracted separately | part of the battery is reserved for the <a href="#">PPA</a> product | customer premium exceeds <a href="#">opportunity cost</a> and risk boundary is clear |
| Combined hybrid <a href="#">PPA</a> | renewable energy and <a href="#">BESS</a> are packaged into one customer product                            | larger share of flexibility is locked to the customer               | buyer values firmness enough to pay for it and evidence/control rights are explicit  |

## Worked Solution

| Route                                             | Expected Value        | Downside Value              | Financeable Value In This Scenario         | Main <a href="#">Constraint</a> |
|---------------------------------------------------|-----------------------|-----------------------------|--------------------------------------------|---------------------------------|
| Full merchant                                     | 1.70 million EUR/year | 0.70 million EUR/year       | 0.34 million EUR/year                      | Volatile and weakly financeable |
| Floor contract                                    | 1.20 million EUR/year | 0.90 million EUR/year       | 0.90 million EUR/year                      | Gives up some upside            |
| Shaped <a href="#">PPA</a> plus residual merchant | 1.55 million EUR/year | depends on residual firming | 0.75 million EUR/year plus residual credit | Ties up control rights          |

The full merchant route has the highest expected value, but the lowest financeable value. The floor contract has lower expected value but stronger downside and lender readability. The

shaped PPA route is attractive only if the PPA premium is high enough to compensate for lost merchant optionality.

### Numerical Intuition

The opportunity cost of using battery capacity for the shaped PPA is:

$$\begin{aligned} & \text{full merchant expected value} - \text{residual merchant value} \\ &= 1.70 \text{ million} - 0.80 \text{ million} \\ &= 0.90 \text{ million EUR/year} \end{aligned}$$

The shaped PPA pays 0.75 million EUR/year for that flexibility. On expected value alone, it is short by 0.15 million EUR/year before considering customer value, contract tenor, and financeability.

## Decision Memo

Recommendation: contract a portion of the asset under a floor-like structure and keep a meaningful residual slice merchant. Do not allocate battery capacity to the shaped PPA unless the PPA premium rises or the PPA materially improves project financeability.

The binding constraint is control-right allocation. Every MW reserved for a customer profile is a MW unavailable for reserve capacity, arbitrage, or response to a new market product. The term sheet should therefore state the priority order for reserve obligations, customer shape delivery, merchant arbitrage, degradation limits, insurance-cycle limits, minimum state of charge, and emergency override.

The commercial team should price shaped PPA flexibility at least at its opportunity cost, not at the accounting cost of the battery. If the customer cannot pay the opportunity cost, the better product is a narrower shaped PPA plus a separate floor or optimizer agreement for the battery.

The decision pack should show two values side by side: expected value and financeable downside value. A route that maximizes expected merchant upside may still be the wrong first

financing structure if it leaves the project with weak debt sizing, high collateral exposure, or no lender-readable revenue floor.

## Assumption Flip

The recommendation changes if merchant revenues saturate and the expected merchant stack falls by 40%.

$$1.70 \text{ million EUR/year} \times 60\% = 1.02 \text{ million EUR/year}$$

In that world, the 0.90 million EUR/year floor becomes much more attractive, and the shaped [PPA](#) premium may clear the opportunity-cost test.

## Failure Mode To Avoid

Use last year's reserve revenue, an arbitrage forecast, and a shaped [PPA](#) premium as three additive upside lines, then haircut the total for prudence.

The same failure appears when the stack ignores policy constraints. If the insurance contract allows fewer cycles, different duty cycles, or fewer operating modes than the optimizer assumes, part of the stack is unavailable unless the owner changes coverage or changes operation.

That still double-counts flexibility. The same MW, MWh, state of charge, control rights, cycle budget, and grid connection must serve every product. Revenue stacking is constrained co-optimization, not spreadsheet addition.

## Links Back To The Lectures

- [Frequency And Reserves](#)
  - [Batteries As Time-Shifting Assets](#)
  - [BESS Revenue Stack](#)
  - [Dispatch Optimization](#)
  - [Valuation Under Uncertainty](#)
  - [Batteries After Ancillary-Service Saturation](#)
-

# Case 5. Balance-Responsibility And Balancing-Service Performance Audit

References: [23] [18] [24] [25] [26] [27]

A trading or optimization partner can execute the market interface, but the asset owner still needs evidence before it changes fees, mandates, incentives, or partners. The hard part is attribution. A monthly revenue gap may come from market conditions, missing data, asset constraints, forecast error, risk limits, or dispatch choices. Partner action is justified only after the evidence separates those causes.

## Decision Question

Should the asset owner change partner terms now, require evidence remediation first, or treat the gap as market and constraint noise?

## Situation

A renewable IPP outsources balance responsible party (BRP) and balancing service provider (BSP) execution work. The partner sends monthly revenue reports. The asset owner wants to know whether the partner created value, leaked value, or operated within realistic constraints before opening a commercial renegotiation.

The partner contract allows performance review, but the current data annex is weak. Forecast files, bid files, schedule revisions, optimizer recommendations, manual overrides, meter data, activation records, and settlement corrections are delivered inconsistently.

## Scenario Data

These are scenario numbers for one month. The revenue row is already net of imbalance, curtailment, and reserve effects. The lower rows are raw evidence inputs for attribution, not extra lines to add or subtract from revenue.

| Metric                         | Actual Partner Result | Feasible Benchmark |
|--------------------------------|-----------------------|--------------------|
| Net energy and reserve revenue | 5.90m EUR             | 6.40m EUR          |
| <u>Imbalance</u> cost          | -0.42m EUR            | -0.31m EUR         |

| Metric                                | Actual Partner Result | Feasible Benchmark |
|---------------------------------------|-----------------------|--------------------|
| Curtailed or unavailable energy value | -0.18m EUR            | -0.17m EUR         |
| Data completeness                     | 92%                   | 99% required       |
| Forecast files delivered on time      | 88%                   | 98% required       |
| Bids matching agreed risk limits      | 100%                  | 100% required      |

The feasible benchmark uses only information available before gate closure, actual grid and asset limits, contracted control rights, agreed risk limits, and market products the partner was allowed to access.

Raw attribution pack:

| Evidence Item                      | Observation                                                            | Initial EUR Impact |
|------------------------------------|------------------------------------------------------------------------|--------------------|
| Missing or late data               | forecast and bid files missing for several high-spread intervals       | 0.12m              |
| Grid and asset constraints         | export limit and outage periods reduced feasible dispatch              | 0.08m              |
| Forecast error inside agreed range | weather-driven production miss within tolerance band                   | 0.09m              |
| Risk-limit conservatism            | partner stayed inside owner-approved risk limits during volatile hours | 0.10m              |
| Residual bidding/dispatch gap      | unexplained difference after known constraints                         | 0.11m              |

Evidence quality flags:



| Required Evidence                | Status                           | Consequence                                      |
|----------------------------------|----------------------------------|--------------------------------------------------|
| Forecast versions and timestamps | incomplete                       | cannot prove what was knowable                   |
| Bid submissions and revisions    | partial                          | cannot fully test partner action                 |
| Optimizer recommendations        | missing for 18% of intervals     | cannot separate tool advice from manual override |
| Asset availability and outages   | complete                         | physical constraints can be attributed           |
| Meter and settlement data        | provisional, corrections pending | revenue result may move                          |
| Contract rights and risk limits  | complete                         | feasible benchmark boundary is known             |

## Tasks

1. Calculate the headline gap.
2. Attribute the gap using the raw evidence.
3. Decide which amount can be used for partner action now.
4. Draft a scorecard and governance recommendation.

## Worked Solution

Headline gap:

```
benchmark_revenue - actual_revenue
= 6.40m EUR - 5.90m EUR
= 0.50m EUR
```

Attribution:

| Driver                             | EUR Impact   | Governability                            | Can Support Partner Action Now?      |
|------------------------------------|--------------|------------------------------------------|--------------------------------------|
| Missing or late data               | 0.12m        | shared governance / contract data rights | yes, for data remediation            |
| Grid and asset constraints         | 0.08m        | mostly non-controllable                  | no                                   |
| Forecast error inside agreed range | 0.09m        | model improvement                        | no, unless tolerance is renegotiated |
| Risk-limit conservatism            | 0.10m        | owner policy choice                      | no, unless mandate changes           |
| Residual bidding/dispatch gap      | 0.11m        | partner execution candidate              | only after evidence file is complete |
| <b>Total</b>                       | <b>0.50m</b> |                                          |                                      |

The only clean commercial action this month is data remediation. The 0.11m EUR residual may be partner execution leakage, but the missing forecast, bid, and optimizer files make a fee claim premature.

## Numerical Intuition

The headline gap overstates the partner-action amount.

```
residual_bidding_gap_share = 0.11 / 0.50 = 22%
data_and_observability_share = 0.12 / 0.50 = 24%
```

Nearly a quarter of the gap is an evidence problem, and only about a fifth is a candidate execution problem. A strong scorecard prevents the owner from negotiating a 0.50m EUR accusation when the supported execution claim is closer to 0.11m EUR and still evidentially weak.

## Evidence Required Before Partner Action

The owner should require raw files before using the scorecard for fee clawback, termination, incentive reset, or mandate reduction.

| Evidence Layer             | Required Files Or Data                                                                  | What It Separates                                        |
|----------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------|
| Forecast and schedules     | forecast versions, schedules, revisions, timestamps, gate-closure state                 | model error versus late information                      |
| Bids and dispatch          | submitted bids, accepted bids, activations, optimizer recommendations, manual overrides | partner choice versus market acceptance                  |
| Asset and grid constraints | availability, outages, import/export limits, reserve commitments                        | controllable execution versus physical <u>constraint</u> |
| Meter and settlement       | metered volumes, <u>imbalance</u> settlement, reserve settlement, corrections           | economic result versus provisional estimate              |

| Evidence Layer               | Required Files Or Data                                                          | What It Separates                                 |
|------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------|
| Contract rights              | data rights, control rights, risk limits, permitted products                    | what the partner was allowed to do                |
| Credit and <u>collateral</u> | <u>collateral</u> calls, credit limits, settlement-bank or counterparty notices | revenue result versus liquidity <u>constraint</u> |

This mirrors the logic in Nordic settlement governance: performance and imbalance outcomes depend on reported data, settlement quantities, participant obligations, and corrections. A private partner scorecard should be at least as disciplined about raw evidence.

## Scorecard

Use a scorecard that separates result, evidence, and attribution.

| Scorecard Area                    | Month Result               | Status | Action                                                     |
|-----------------------------------|----------------------------|--------|------------------------------------------------------------|
| Revenue versus feasible benchmark | -0.50m EUR                 | watch  | track with attribution; no fee action on headline gap      |
| Data completeness                 | 92% versus 99% requirement | fail   | remediation plan with dated file-delivery obligations      |
| Forecast timeliness               | 88% versus 98% requirement | fail   | daily timestamp feed and exception report                  |
| Risk-limit compliance             | 100%                       | pass   | keep current limits until management reviews risk appetite |

| Scorecard Area                | Month Result | Status     | Action                                                     |
|-------------------------------|--------------|------------|------------------------------------------------------------|
| <u>Constraint attribution</u> | 0.08m EUR    | accepted   | exclude from partner execution claim                       |
| Residual execution candidate  | 0.11m EUR    | unresolved | retest after complete raw files and settlement corrections |

The scorecard should be repeated for at least three comparable months before commercial penalties are applied, unless the contract already defines automatic data-delivery breaches.

## Decision Memo

Recommendation: do not renegotiate fees or accuse the partner based on the 0.50m EUR headline gap. Issue a formal data-rights and reporting remediation notice, agree the feasible-benchmark methodology, and put the partner on a three-month evidence watch.

Immediate actions:

- amend or enforce the data annex for forecast, schedule, bid, dispatch, optimizer, meter, activation, and settlement files
- require timestamped delivery and exception reporting
- freeze the feasible-benchmark method before seeing next month's results
- review whether owner-approved risk limits are too conservative for volatile periods
- retest residual bidding/dispatch gaps only where evidence is complete

Partner commercial action becomes supportable if, for three comparable months:

- data completeness meets the 99% requirement
- forecast timeliness meets the 98% requirement
- settlement corrections are reconciled
- the residual bidding/dispatch gap remains material
- the gap occurs in intervals where action was feasible and permitted

## Assumption Flip

The recommendation changes if complete raw files show that the partner ignored feasible profitable bids, violated the agreed strategy, or failed to act inside clear market windows while limits and asset constraints allowed action.

In that case, the 0.11m EUR residual becomes an execution claim rather than an unresolved attribution bucket, and the owner can move to fee adjustment, incentive redesign, tighter mandate, or replacement process.

## Failure Mode To Avoid

Use a perfect-hindsight dispatch and attribute the full 0.50m EUR gap to the partner.

Perfect hindsight can estimate an upper bound, but it cannot govern a BRP/BSP relationship. The governance benchmark must use forecasts, market prices, liquidity, grid limits, asset status, control rights, and risk limits available at the time.

## Links Back To The Lectures

- [Cast Of Characters](#)
  - [Market Clock](#)
  - [Forecast Error And Imbalance](#)
  - [Frequency And Reserves](#)
  - [Forecasting Value And Error](#)
  - [Shadow Trading And Partner Scorecards](#)
  - [Analytical Capability Roadmap](#)
-

## Case 6. Data-Centre 24/7 Product Screen

References: [1] [4] [28] [5] [2]

Large electricity buyers increasingly care about both when clean energy is produced and how many clean MWh are bought over a year. The seller must separate physical supply, financial settlement, certificate evidence, residual firming, credit, and control rights. The decision is whether a renewable [IPP](#) should offer a 24/7-style product, narrow the promise, partner, or walk away.

### Decision Question

Should the seller offer hourly matched clean power, a narrower shaped product, a partnered offer, or no-bid?

### Situation

A data-centre buyer requests a clean-power product for a flat 40 MW load. The buyer asks for high [hourly matching](#), budget stability, and clear evidence treatment.

The seller has solar assets, access to a battery project, and a trading partner. It does not own enough diverse clean generation to self-supply every hour.

### Scenario Data

These are scenario assumptions.

| Item                                                 | Value          |
|------------------------------------------------------|----------------|
| Buyer load                                           | 40 MW flat     |
| Annual demand                                        | 350.4 GWh/year |
| Seller-controlled clean supply                       | 260 GWh/year   |
| Annual clean-energy coverage                         | 74%            |
| <a href="#">Hourly matching</a> before extra firming | 58%            |

| Item                                              | Value       |
|---------------------------------------------------|-------------|
| Target <u>hourly matching</u> for premium product | 85%         |
| Residual stress-hour shortfall                    | 22 MWh/h    |
| Stress hours                                      | 300 h/year  |
| Firming premium in stress hours                   | 150 EUR/MWh |
| Buyer premium for stronger <u>hourly matching</u> | 4 EUR/MWh   |

## Tasks

1. Calculate the annual buyer demand.
2. Calculate stress-hour firming-at-risk.
3. Compare the buyer premium with the stress exposure.
4. Define the product boundary and evidence standard.
5. Recommend the product boundary.

## Worked Solution

Annual demand:

$$40 \text{ MW} \times 8,760 \text{ h/year} = 350,400 \text{ MWh/year}$$

Stress-hour firming-at-risk:

$$22 \text{ MWh/h} \times 300 \text{ h/year} \times 150 \text{ EUR/MWh} \\ = 990,000 \text{ EUR/year}$$

Buyer premium:

$$350,400 \text{ MWh/year} \times 4 \text{ EUR/MWh} \\ = 1,401,600 \text{ EUR/year}$$



The premium covers the illustrated stress-hour procurement cost before collateral, credit, evidence handling, operational costs, and margin. The case still needs careful structuring before rejection or approval.

## Product Boundary

The product should be split into claims that can be evidenced and controlled.

| Claim Or Service                 | Required Evidence Or Control                                                                                                  | Seller Exposure If Vague                                              |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Annual clean-energy coverage     | generation volumes, certificates, delivery period, geography, allocation rules                                                | buyer receives a green volume claim but not an hourly product         |
| <u>Hourly matching</u> threshold | hourly load, hourly clean supply, storage dispatch, residual procurement records                                              | seller may be judged against a standard it did not price              |
| Residual firming                 | cap, index, pass-through, partner supply, or reopener for hard hours                                                          | uncapped procurement exposure in scarce hours                         |
| Budget stability                 | settlement reference, volume tolerance, <u>collateral</u> and credit terms                                                    | clean-energy product becomes an unpriced financial hedge              |
| Certificate treatment            | certificate vintage, cancellation, location, matching period, audit trail                                                     | environmental claim can fail after the commercial deal is signed      |
| Buyer flexibility                | whether uninterruptible power supply (UPS), cooling, workload shifting, or on-site generation can be used and who controls it | load flexibility is assumed in the model but unavailable in operation |

For a data-centre buyer, flexibility resources can be valuable, but reliability constraints come first. Do not count buyer-side flexibility as seller supply unless the buyer grants defined availability, control, telemetry, and settlement rights.

| Numerical Intuition                                                       |                       |                                           |
|---------------------------------------------------------------------------|-----------------------|-------------------------------------------|
| Annual coverage can look acceptable while <u>hourly matching</u> is weak. |                       |                                           |
| Metric                                                                    | Result                | Why It Matters                            |
| Annual clean-energy coverage                                              | 74%                   | Useful but not enough for timing exposure |
| <u>Hourly matching</u> before firming                                     | 58%                   | Shows residual hourly procurement need    |
| Stress-hour <u>firming-at-risk</u>                                        | 0.99 million EUR/year | Main seller exposure in hard hours        |
| Buyer premium                                                             | 1.40 million EUR/year | Commercial headroom before other costs    |

## Decision Memo

Recommendation: offer a partnered hourly-matching product only if the contract defines the evidence standard, residual firming boundary, settlement method, and fallback procurement rule. Otherwise offer a narrower shaped product and leave the buyer with the residual claim.

The seller should not promise a broad 24/7 clean-energy claim without the applicable accounting and certificate framework in the contract. The commercial product can still be valuable if it says exactly what is being delivered: hourly matched supply to a defined threshold, financial settlement against a defined reference, and residual procurement above a defined boundary.

The recommended offer is therefore a product ladder:

| Offer                             | Seller Commitment                                                                            | Buyer Benefit                  | When To Use                                                    |
|-----------------------------------|----------------------------------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------|
| As-produced plus certificates     | clean energy volume and evidence                                                             | lowest complexity              | buyer accepts timing exposure                                  |
| Shaped clean product              | defined hourly shape with residual boundary                                                  | budget and profile improvement | buyer values shape but not full hourly claim                   |
| Partnered hourly-matching product | threshold-based <a href="#">hourly matching</a> with named firming partner and evidence pack | stronger 24/7-style claim      | premium covers firming, evidence, credit, and operational cost |
| No-bid on broad claim             | seller refuses uncapped 24/7 language                                                        | avoids mispriced obligation    | product boundary cannot be made auditable                      |

## Assumption Flip

The recommendation changes if stress-hour shortfall rises to 40 MWh/h.

$$\begin{aligned}
 &40 \text{ MWh/h} \times 300 \text{ h/year} \times 150 \text{ EUR/MWh} \\
 &= 1,800,000 \text{ EUR/year}
 \end{aligned}$$

That exceeds the buyer premium before other costs. The seller should then require a higher premium, a partner portfolio, a narrower matching threshold, or a no-bid decision.

## Failure Mode To Avoid

Offer the product as a premium clean-power contract because annual clean-energy coverage is high, while leaving hourly matching rules, certificate evidence, and residual firming treatment to later documentation.

That decision sells the label before defining the product. [Annual matching](#) can hide the hours that drive firming cost, evidence disputes, and residual procurement exposure. A flat load

needs a time-indexed product boundary alongside the annual energy total.

## Links Back To The Lectures

- [Load Is Not Energy](#)
  - [Market Clock](#)
  - [Solar Shape](#)
  - [PPA Structure Catalog](#)
  - [PPA Pricing](#)
  - [Forecasting](#)
  - [Portfolio Modeling And Hedging](#)
  - [Trading Risk, Credit, Liquidity, And Controls](#)
  - [Future Of PPAs](#)
  - [Data Centers And 24/7 Clean Energy](#)
-

# Case 7. Year-One Energy-Management Operating Model

References: [6] [27] [29] [26] [1] [5]

An independent power producer (IPP) becomes an energy manager when it decides which revenue and risk decisions it must own, which execution activities it can outsource, what raw data it has rights to inspect, and which controls stop a model from becoming an unmanaged production process. This case forces the year-one resource allocation.

## Decision Question

Given limited budget, headcount, partner data gaps, and several competing value pools, what should the company build, buy, outsource, audit, or postpone in year one?

## Situation

The company has operating solar assets, a growing development pipeline, several structured PPA discussions, and first battery projects entering commercial design. Trading execution and balancing are mostly outsourced. Partner reporting is useful but incomplete. Management wants a credible year-one operating model without approving a large desk or full energy trading and risk management (ETRM) implementation.

The operating model must support four decisions:

- shaped PPA pricing
- BESS sizing and dispatch value
- BRP/BSP and optimizer oversight
- portfolio exposure and risk escalation

## Scenario Data

These are scenario assumptions.

| Item                                         | Value              |
|----------------------------------------------|--------------------|
| Operating renewables                         | 600 MW             |
| Development pipeline under commercial review | 1.8 gigawatts (GW) |

| Item                                             | Value                                 |
|--------------------------------------------------|---------------------------------------|
| First <u>BESS</u> projects                       | 3 projects                            |
| Annual <u>PPA</u> offers under active discussion | 700 GWh/year                          |
| Outsourced execution volume                      | 500 GWh/year                          |
| Current internal analytics team                  | 1 accountable owner + 1 analyst       |
| Additional hire allowed                          | 0.5 data engineer contractor          |
| Year-one budget                                  | 900,000 EUR                           |
| External execution model                         | <u>BRP/BSP</u> and optimizer partners |
| Management <u>constraint</u>                     | no internal trading desk in year one  |

#### Partner data-right gaps:

| <u>Data Right</u>                              | Current Status                                    | Operating Consequence                             |
|------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| Forecast versions and timestamps               | partial                                           | hard to judge what was knowable                   |
| Bid and schedule revisions                     | partial                                           | partner actions cannot be fully reconstructed     |
| Optimizer recommendations                      | limited                                           | manual override versus model decision is unclear  |
| Meter and settlement files                     | available with delay                              | monthly attribution lags                          |
| Reserve <u>activation</u> and imbalance detail | inconsistent                                      | <u>revenue stack</u> cannot be reconciled cleanly |
| Control-right map                              | contract language exists, operational map missing | dispatch priority conflicts are hard to prove     |

Value pools:

| Value Pool                                    | Scenario Value At Stake | Evidence Quality              |
|-----------------------------------------------|-------------------------|-------------------------------|
| Shaped <a href="#">PPA</a> risk pricing       | 560,000 EUR/year        | medium-high                   |
| BESS design and <a href="#">revenue stack</a> | 750,000 EUR/year        | medium                        |
| Partner leakage reduction                     | 350,000 EUR/year        | low until data rights improve |
| Portfolio risk and hedge decisions            | 250,000 EUR/year        | medium                        |

Control failures observed:

- commercial offers use inconsistent shape, basis, certificate, and residual-firming assumptions
- [BESS](#) cases double-count merchant and PPA use of the same MWh in some sensitivities
- partner reports show results without complete raw files
- management sees [portfolio exposure](#) after settlement rather than before decisions
- model versions and assumption owners are unclear

## Tasks

1. Allocate the 900,000 EUR budget across year-one capabilities.
2. Assign headcount ownership and name what remains outsourced.
3. Prioritize a 90-day plan and a 12-month roadmap.
4. State which partner data-right gaps block partner action.
5. Define kill criteria for the roadmap.

## Worked Solution

Year-one resource allocation:

| Capability                                      | Build, Buy, Outsource, Audit, Or Postpone | Budget      | Owner                                | Reason                                           |
|-------------------------------------------------|-------------------------------------------|-------------|--------------------------------------|--------------------------------------------------|
| Data contracts and reconciliation               | build/audit                               | 130,000 EUR | accountable owner + data engineer    | unlocks all other controls                       |
| PPA <a href="#">quote stack</a> and risk map    | build                                     | 160,000 EUR | accountable owner + commercial lead  | shaped offers need consistent risk pricing       |
| <a href="#">BESS</a> sizing and dispatch screen | build with optimizer input                | 140,000 EUR | analyst + investment owner           | IC decisions are live                            |
| Partner data-right remediation                  | audit/renegotiate                         | 120,000 EUR | accountable owner + legal/commercial | raw evidence is missing                          |
| Portfolio risk pack                             | build                                     | 110,000 EUR | analyst + risk owner                 | management needs one exposure view               |
| Forecast/vendor evaluation layer                | buy inputs, build evaluation              | 90,000 EUR  | analyst                              | value-weighted errors support multiple decisions |
| Feasible benchmark pilot                        | build small pilot                         | 50,000 EUR  | accountable owner                    | method first, automation later                   |
| Contingency and                                 | buy                                       | 100,000 EUR | steering group                       | avoids brittle models and                        |



| Capability         | Build, Buy, Outsource, Audit, Or Postpone | Budget             | Owner | Reason          |
|--------------------|-------------------------------------------|--------------------|-------|-----------------|
| independent review |                                           |                    |       | weak validation |
| <b>Total</b>       |                                           | <b>900,000 EUR</b> |       |                 |

Postpone:

- internal trading desk
- full [ETRM](#) implementation
- automated benchmark platform across all assets
- proprietary weather model

Outsource:

- BRP/BSP [market access](#)
- 24/7 dispatch execution
- base weather/price forecast feeds
- some legal/accounting/certificate advice

Own internally:

- [PPA](#) risk allocation and quote logic
- [BESS](#) investment assumptions
- data-right requirements
- feasible benchmark methodology
- monthly portfolio risk view
- [model register](#) and assumption ownership

## Forced Tradeoff

Building a polished partner scorecard immediately would spend money before the evidence layer is ready. Partner data-right gaps make a full scorecard expensive and weak. The stronger year-one allocation funds the data annex and only a narrow benchmark pilot.

full\_scorecard\_without\_data\_rights = high spend + weak attribution  
data\_rights\_first + small pilot = slower automation + stronger governance

The PPA [quote stack](#) and BESS screen receive more year-one funding because live commercial and IC decisions can use them with available internal data. Partner commercial action waits for raw evidence.

## First 90 Days

| Month | Deliverable                                             | Decision It Enables                                                                                         |
|-------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 1     | data-rights inventory and gap notice to partners        | shows what can be measured, audited, and renegotiated                                                       |
| 1     | <a href="#">model register</a> and assumption owner map | prevents uncontrolled spreadsheets                                                                          |
| 2     | shaped-PPA <a href="#">quote stack</a> v1               | quote, redesign, no-bid, or pass-through decision                                                           |
| 2     | <a href="#">BESS</a> duration/revenue-stack screen v1   | 1h/2h/4h, grid-right, <a href="#">degradation</a> , and control-right decisions                             |
| 3     | first monthly portfolio risk pack                       | management view of price, shape, basis, <a href="#">imbalance</a> , credit, liquidity, and partner exposure |
| 3     | <a href="#">partner benchmark</a> methodology note      | agrees feasible benchmark rules before automation                                                           |

The first 90 days should produce usable decision artifacts, not a finished system.

## Twelve-Month Roadmap

| Quarter | Milestone                                         | Acceptance Test                                                                                     |
|---------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Q1      | common <u>data contract</u><br>inventory          | every critical feed has source, owner, timestamp, unit, granularity, validation rule, and gap owner |
| Q2      | PPA and <u>BESS</u> models used in live decisions | at least one quote or IC recommendation changes because of the model                                |
| Q2      | risk pack live                                    | management uses it for limits, hedge questions, partner escalation, or deal approval                |
| Q3      | forecast evaluation layer                         | day-ahead, intraday, <u>BESS</u> , and partner forecast errors are value-weighted                   |
| Q3      | partner data annex amended or escalation logged   | missing raw evidence has a commercial owner                                                         |
| Q4      | benchmark pilot completed                         | one asset class or partner has feasible attribution using time-available data                       |

## Risk Pack

The recurring risk pack should be short and decision-linked.

| Section             | Minimum Content                                                    | Escalation Trigger              |
|---------------------|--------------------------------------------------------------------|---------------------------------|
| Executive exposure  | price, shape, basis, <u>merchant tail</u> , and liquidity summary  | exposure outside approved range |
| <u>PPA pipeline</u> | <u>quote stack</u> , residual firming, certificates, credit, basis | unpriced risk in term sheet     |

| Section                       | Minimum Content                                                                        | Escalation Trigger                    |
|-------------------------------|----------------------------------------------------------------------------------------|---------------------------------------|
| <u>BESS</u> and dispatch      | <u>revenue stack</u> , state-of-charge constraints, degradation, reserve/PPA conflicts | double-counted MW/MWh or cycle budget |
| Partner scorecard             | data completeness, forecast timeliness, feasible benchmark gap                         | raw evidence below requirement        |
| Settlement and reconciliation | meter, <u>imbalance</u> , reserve, invoice exceptions                                  | unreconciled material variance        |
| Credit and <u>collateral</u>  | counterparty exposure, <u>collateral</u> calls, guarantees                             | liquidity stress or credit breach     |
| Model changes                 | version changes, assumption moves, validation exceptions                               | unapproved model used in decision     |

The pack is a control only if management uses it to approve, stop, redesign, hedge, escalate, or request evidence.

## Data Rights Before Partner Action

Partner action is blocked where raw evidence is missing.

| Action                    | Evidence Required First                                                              |
|---------------------------|--------------------------------------------------------------------------------------|
| fee reduction or clawback | complete bids, schedules, dispatch recommendations, activations, and settlement data |
| mandate change            | proof that current limits caused avoidable loss or excess risk                       |
| optimizer replacement     | evidence that feasible recommendations were materially worse than benchmark          |

| Action                              | Evidence Required First                                                                       |
|-------------------------------------|-----------------------------------------------------------------------------------------------|
| <a href="#">BRP/BSP replacement</a> | repeated residual execution gap after data, constraints, forecast, and risk-limit attribution |
| incentive redesign                  | clear split between market conditions, constraints, and controllable partner decisions        |

Until those files exist, the company can demand data remediation and better reporting. It should be cautious about claiming execution leakage.

## Kill Criteria

| Workstream                      | Continue If...                                                       | Kill Or Redesign If...                                                                         |
|---------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| PPA <a href="#">quote stack</a> | used in live offers and changes price or structure                   | commercial team bypasses it or outputs cannot explain residual risk                            |
| <a href="#">BESS</a> screen     | changes IC decisions or preserves valuable design option             | dispatch value cannot be tied to grid rights, <a href="#">degradation</a> , and control rights |
| Data remediation                | partner completeness and timeliness improve against dated milestones | contract rights remain blocked after escalation                                                |
| Risk pack                       | management takes decisions from it                                   | it becomes a passive report                                                                    |
| Forecast layer                  | <a href="#">value-weighted error</a> improvement exceeds cost        | updates arrive too late for action                                                             |
| Benchmark pilot                 | attribution uses time-available data and settlement reconciliation   | it relies on hindsight or missing files                                                        |

## Decision Memo

Recommendation: build a small control-oriented energy-management function in year one while keeping market execution outsourced.

Approve the full 900,000 EUR allocation shown above. The first year should produce a PPA [quote stack](#), BESS design screen, data-right remediation, monthly portfolio risk pack, forecast evaluation layer, and a narrow feasible-benchmark pilot. Do not approve an internal trading desk or full ETRM implementation until data rights, independent settlement checks, risk limits, and evidence-based partner attribution are working.

The operating model is deliberately constrained:

- one accountable owner coordinates decisions, partner governance, and steering
- one analyst owns models, risk pack production, and value-weighted forecast analysis
- the half-time data engineer builds data contracts, reconciliation checks, and repeatable feeds
- partners continue to execute, but their raw evidence obligations become explicit

## Assumption Flip

The recommendation changes if partners refuse or cannot provide forecast, bid, schedule, dispatch, meter, [activation](#), and settlement files on a reliable timeline.

In that case, shift more budget from model expansion to contract remediation, data interfaces, and replacement-option analysis. Building more models before the company can observe partner decisions would create false control.

## Failure Mode To Avoid

Internalize market execution before the company has data rights, independent settlement checks, feasible benchmarks, risk limits, and escalation rules.

That sequence creates operational and control risk. Execution capability only helps when the organization can see the decisions being made, attribute outcomes, enforce limits, and settle disputes with evidence.

## Links Back To The Lectures

- Forecasting Value And Error
  - Shadow Trading And Partner Scorecards
  - Analytical Capability Roadmap
  - Artificial Intelligence And Machine Learning In Energy Management
  - Build, Buy, Outsource Trading Layer
  - Next-Generation IPP Operating Model
-

# Glossary

An alphabetical reference for the main terms used in the course.

## Actionability factor

Fraction of a model improvement that can become value after information timing, market access, liquidity, control rights, and constraints.

## Actionability-adjusted uplift

Raw model or capability uplift after reducing for timing, tradable volume, liquidity, asset constraints, control rights, and organizational adoption.

## Activation

Automatic or instructed conversion of reserve capability into physical response.

## aFRR

Automatic Frequency Restoration Reserve: reserve automatically activated to restore frequency/control area balance after disturbance.

## Aggregator

Entity that combines distributed assets or loads so they can participate commercially or technically as a portfolio.

## Allocated capacity

Grid capacity assigned or reserved for a connection or network user under a specific process or agreement.



### **Analytical engine**

Connected set of models, data flows, controls, and decision rights that turns market and asset information into repeatable commercial decisions.

### **Annual matching**

Comparing clean-energy procurement and consumption over a year.

### **Apparent power**

Combined active and reactive loading of AC equipment, commonly represented as  $S$  where  $S^2 = P^2 + Q^2$ .

### **As-produced PPA**

PPA where buyer takes generation as produced, inheriting shape/profile exposure.

### **Assumption-flip test**

Search for the smallest plausible change in an assumption that would reverse a recommendation.

### **Augmentation**

Addition or replacement of battery equipment to restore or maintain usable capacity.

### **Back office**

Function responsible for confirmations, settlements, invoicing, contract administration, reporting, and operational control records.

### **Backtest**

Historical simulation of model or policy performance using only information available before each decision time.

**Balance obligation**

Obligation to plan for balance and financially settle imbalances under market rules.

**Balancing capacity**

MW held available for balancing service under TSO product rules, whether or not activation occurs.

**Balancing energy**

Energy delivered or absorbed when balancing bids or reserves are activated.

**Balancing market**

Real-time or near-real-time mechanism used by the TSO to maintain system balance.

**Bankable revenue**

Revenue treated as dependable enough for financing, depending on contract, counterparty, tenor, and downside protection.

**Baseload PPA**

PPA with a flat delivery profile over a defined period.

**Basis risk**

Mismatch between the exposure being hedged and the instrument/reference used to hedge it.

**Battery duration**

Battery energy capacity divided by power rating, usually expressed in hours.

**Behavioral market power**

Actual conduct that uses market-power potential through bidding, withholding, scheduling, or dispatch choices.

### **BESS**

Battery Energy Storage System.

### **BRP**

Balance Responsible Party: entity financially responsible for imbalances at a delivery point or portfolio.

### **BSP**

Balance Service Provider: entity providing balancing services to the TSO.

### **Calculation period**

Time period over which a reference price or settlement amount is calculated.

### **Calendar aging**

Battery degradation associated with time, temperature, and state-of-charge conditions, even without heavy cycling.

### **Cannibalization**

Value erosion when similar generators produce at the same time and depress prices during their own production hours.

### **Cap**

Contractual maximum price or value above which upside is limited or shared, depending on structure.

### **Capability priority score**

Heuristic ranking of a capability by value at stake, controllability, observability, urgency, cost, and risk.

### **Capability-contract matrix**

Build/buy/outsource table that separates internal judgment, tools, outsourced services, and non-negotiable control/data rights.

### **Capacity credit**

Dependable capacity contribution of a resource divided by its nameplate capacity.

### **Capacity factor**

Actual energy produced over a period divided by energy that would be produced at full nameplate output over that period.

### **Capacity remuneration mechanism**

Market or regulatory mechanism that pays for capacity, availability, or reliability contribution in addition to energy.

### **Capture price**

Generation-weighted average market price received by an asset or technology.

### **Capture rate**

Capture price divided by average spot price for the same market/period.

### **Cash available for debt service**

Cash flow available to pay scheduled debt service after operating revenue, operating costs, and allowed deductions/reserves according to the model or financing terms.

### **Cash Flow at Risk**

Liquidity metric focused on potential cash demands versus available sources over a horizon.

### **Certificate contract price**

Price or embedded value assigned to certificates such as GOs or EECS certificates in a PPA.

### **CfD**

Contract for Difference: financial contract settling the difference between a strike/reference price and a market/reference price.

### **Closed-loop latency ratio**

Feedback latency divided by decision cadence; high values mean the organization learns slower than it acts.

### **Collateral**

Value posted to reduce credit exposure between counterparties or clearing participants.

### **Congestion rent**

Value associated with a price difference across a constrained transmission interface multiplied by feasible flow, in simplified market-design examples.

### **Connection agreement**

Agreement governing how an asset connects to the grid, including technical, capacity, cost, and operational conditions.

### **Connection charge**

Charge, often one-off, covering part or all of connecting a new user or upgrading a

connection.

### **Constraint**

Physical, market, contractual, or operational limit a model or decision must obey.

### **Construction contribution**

One-off payment or cost contribution for grid works required to connect or serve a network user.

### **Contract quantity**

Electricity quantity specified by a PPA or related contract, such as all metered output, part of metered output, or a fixed delivery schedule.

### **Control right**

Authority to decide, instruct, or veto scheduling, bidding, dispatch, curtailment, hedging, or reserve actions.

### **Cost of capital**

Required return for the capital supplied to a project, including debt and equity costs.

### **Counterparty credit risk**

Risk that the counterparty to a contract does not pay, deliver, or otherwise perform as required.

### **Credit support**

Collateral, guarantee, parent support, letter of credit, or similar instrument used to reduce counterparty credit risk.

### **Cross-border PPA**

PPA where buyer and generation asset are in different national markets or bidding zones, often creating basis and transmission-right questions.

### **Curtailment**

Reduction of output below available generation, for economic, grid, or system-flexibility reasons.

### **Customer value**

Buyer's benefit from a product, such as hedge value, budget certainty, decarbonization claim, hourly alignment, or procurement simplicity; distinct from seller cost.

### **Customer willingness-to-pay**

Maximum price or premium a buyer may rationally accept based on hedge value, budget certainty, decarbonization value, evidence needs, and procurement alternatives.

### **Cycle aging**

Battery degradation associated with charge-discharge use, including throughput, depth of discharge, rate, and operating pattern.

### **Daily risk pack**

Recurring report connecting positions, MtM, market risk, credit exposure, liquidity/collateral, basis/shape, partner execution, limits, data quality, and actions.

### **Data contract**

Definition of a data feed's source, owner, timestamp, unit, granularity, validation rule, and decision use.

### **Data right**

Right to receive data needed to reproduce, audit, and govern decisions.

### **Day-ahead market**

Auction before delivery day where hourly or sub-hourly electricity volumes are cleared for the next day.

### **Debt**

Money lent to a project or company, usually paid before equity and therefore usually cheaper but less tolerant of downside cash-flow weakness.

### **Debt sculpting**

Shaping debt service over time to fit the cash-flow profile a lender is willing to count.

### **Debt sizing**

Process of estimating how much debt a project can support from lender-counted cash flow, coverage requirements, tenor, interest rate, reserves, and amortization structure.

### **Decision latency**

Time between the arrival of useful information and the latest moment a feasible action can still be taken.

### **Decision memo**

Short recommendation document stating the decision, core evidence, retained risk, binding assumptions, owner, next measurement, and assumption-flip test.

### **Decision owner**

Person or function accountable for accepting, rejecting, or overriding a model recommendation or operational action.



**Decision variable**

Quantity the optimization model is allowed to choose.

**Degradation**

Loss of battery capacity or performance over time due to calendar aging and cycling.

**Degradation-adjusted spread**

Discharge price minus charging cost adjusted for efficiency, degradation, fees, tariffs, and opportunity cost.

**Delivery schedule quantity**

Fixed quantity or quantities to be delivered or settled during defined periods under a contract schedule.

**Depth of discharge**

Fraction of stored energy practically used in a discharge.

**Dispatch**

Operational choice of when/how much an asset charges, discharges, generates, or bids.

**Dispatchable**

Controllable by an operator within technical and fuel/resource constraints.

**Distribution**

Lower-voltage network delivery between the transmission system and end users / distributed resources.

**Diversification**

Risk reduction from imperfectly correlated exposures rather than asset count alone.

### **DSCR**

Debt service coverage ratio: cash available for debt service divided by required debt service over a period.

### **DSO**

Distribution System Operator: entity responsible for operating a distribution network.

### **Duration hedge**

Extra battery energy capacity bought partly to handle uncertainty in spreads, delivery obligations, reserve calls, curtailment windows, or grid constraints.

### **Economic withholding**

Offering capacity at a price high enough that it is unlikely to clear, even though it is technically available.

### **Energy**

Quantity accumulated or consumed over time, usually Wh, kWh, MWh, TWh.

### **Energy management**

Capability connecting forecasts, market positions, dispatch, contract obligations, imbalance exposure, partner execution, and portfolio risk.

### **Energy-limited**

Able to provide power only until a stored-energy, fuel, water, or demand-flexibility constraint binds.

### **Energy-management operating system**

Set of recurring decisions, data flows, models, limits, partner controls, feedback loops, and review rhythms that turns assets and contracts into governed commercial action.

### **Energy-only market**

Market design where investment recovery relies mainly on energy and ancillary-service revenues, including scarcity prices.

### **Equity**

Ownership capital that receives residual cash flows after operating costs, debt service, reserves, and other senior claims.

### **Equivalent full cycle**

One full charge-discharge equivalent measured by throughput relative to usable energy capacity; partial cycles can add up to equivalent full cycles depending on the convention used.

### **ETRM**

Energy trading and risk management system used to record, value, monitor, and control trades, positions, limits, settlement, and reporting.

### **Expected shortfall**

Average loss conditional on losses exceeding a selected tail threshold such as VaR.

### **Expected unserved energy**

Expected MWh of demand not served because supply is insufficient.

### **Exposure cube**

Portfolio map that classifies risk by geography, technology, contract, hedge, flexibility,

counterparty, liquidity, and data/control rights.

### **Fault level**

Measure related to system strength and available fault current at a network point.

### **FCR**

Frequency Containment Reserve: reserve intended to contain frequency deviations quickly after an imbalance.

### **Feasible counterfactual**

Simulated action path that could have been taken with contemporaneous information, constraints, market access, risk limits, and control rights.

### **Feedback latency**

Time between a decision and the reliable measurement or settlement data needed to evaluate it.

### **Final position**

Market position carried into delivery after accepted trades, nominations, or adjustments.

### **Financeability lens**

View of project cash flows focused on resilient cash for debt and covenant/downside requirements alongside equity upside.

### **Firming cost**

Cost of transforming variable renewable output into a firmer or more useful delivery shape through storage, portfolio netting, market purchases, hedging, or flexibility.

### **Firming-at-risk**

Stress measure for the cost of residual firming exposure: residual MWh in hard hours multiplied by the relevant firming premium or stress price.

### **Flexibility potential**

Technical or practical ability of a load, generator, storage asset, or portfolio to change injection/withdrawal in response to grid or market needs.

### **Flexible connection agreement**

Connection or network-use arrangement that allows limited, conditional, or interruptible injection or withdrawal, often to accelerate connection or reduce reinforcement needs.

### **Floor**

Contractual minimum price or value below which settlement is protected, depending on structure.

### **Forecast-based dispatch**

Dispatch optimized using information available at decision time.

### **FPA**

Flexibility Purchase Agreement or flexibility-style arrangement; exact meaning must be defined by source/context before use.

### **Front office**

Commercial function that originates, structures, prices, trades, dispatches, or manages market-facing portfolio decisions.

### **Generation adequacy**

Generation-resource component of system adequacy.

### **Governance calendar**

Recurring cadence for position review, partner performance, PPA pricing, portfolio exposure, model assumptions, capability strategy, and investment gates.

### **Granular Guarantee of Origin**

Certificate concept that adds temporal and sometimes locational granularity to energy attribute tracking.

### **Grid tariff**

Charge paid for use of the grid, potentially including fixed, capacity, energy, loss, and reactive-power components.

### **Grid-following inverter**

Inverter that synchronizes to an existing grid voltage/frequency reference.

### **Grid-forming inverter**

Inverter controlled to establish or support voltage/frequency reference behavior more actively.

### **Hedge-quality checklist**

Test of whether a hedge matches the underlying exposure by reference price, volume, shape, tenor, basis, liquidity, collateral, and governance.

### **Hourly matching**

Comparing clean supply and consumption within each hour.

### **Imbalance**

Difference between scheduled/nominated and actual generation or consumption.

### **Incremental duration**

Additional MWh added to a battery design while keeping MW approximately fixed.

### **Incremental duration value**

Additional value from adding MWh to a battery after additional revenue, capex, degradation, tariffs, and opportunity cost.

### **Independent price validation**

Process of checking marks, curves, or valuations independently from the team that originated or owns the position.

### **Inertia**

Resistance of rotating mass to sudden frequency change.

### **Insurance cycle cap**

Insurance-policy limit on annual or lifetime cycling, throughput, depth of discharge, or duty cycle. It can bind even when the physical model and OEM warranty would allow more operation.

### **Insurance operating covenant**

Condition in an insurance policy or endorsement that limits how an asset may be operated while keeping coverage, pricing, or claim eligibility intact. For BESS this may include cycle, throughput, duty-cycle, safety, monitoring, or approved-use restrictions.

### **Intraday market**

Market after day-ahead clearing where participants adjust positions closer to delivery.

### **Investment-case waterfall**

Bridge from gross physical value to equity cash flow that shows capture, curtailment, contract, grid, BESS, opex, debt, and residual effects layer by layer.

### **IPP**

Independent Power Producer: company that owns or operates generation assets and sells electricity or related products without being the incumbent utility.

### **IRR**

Internal rate of return: discount rate that makes project NPV equal zero.

### **Limited grid access**

Connection or access arrangement where injection or withdrawal is constrained under specified conditions.

### **Liquidity at Risk**

Risk measure for liquidity pressure under market, credit, collateral, or trading-liquidity stress.

### **Liquidity risk**

Risk that a firm cannot trade, exit, or fund positions when needed without unacceptable cost or disruption.

### **Load**

Electrical demand at a point, portfolio, zone, or system over time.

### **Load shape**

Pattern of demand across time.

### **Long-duration energy storage**



Storage intended to address longer scarcity periods than short intra-day shifting.

### **Loss charge**

Charge or adjustment intended to recover electrical losses associated with network use.

### **Loss of load expectation**

Expected duration of insufficient supply over a planning horizon.

### **Loss of load probability**

Probability that available resources cannot meet load in a specified period.

### **Margin call**

Demand to post additional collateral after exposure or collateral requirements change.

### **Marginal duration hurdle**

Annualized cost and opportunity-cost threshold that the next MWh of storage energy capacity must clear before extra duration is worth buying or building.

### **Marginal risk contribution**

Incremental stress, variance, liquidity, or concentration added by the next asset, hedge, contract, or partner dependency.

### **Market access**

Ability to participate in a market through required contracts, systems, qualifications, collateral, and operations.

### **Market monitoring**

Monitoring of bids, outages, constraints, behavior, and outcomes to distinguish normal scarcity, opportunity cost, and problematic conduct.

### **Market power**

Ability to profitably influence market price away from the competitive level for a meaningful period.

### **Market risk**

Exposure to losses from movements in prices, spreads, volatility, basis, or correlations.

### **Market time unit**

Time granularity of traded or settled energy, such as hourly or 15-minute periods depending on market and date.

### **Merchant tail**

Post-contracted or partially uncontracted period where project revenue depends more directly on market prices and exposed risks.

### **Merchant upside**

Uncontracted or weakly contracted expected value from market optimization.

### **mFRR**

Manual Frequency Restoration Reserve: reserve manually or centrally activated to restore balance and manage system needs.

### **Middle office**

Independent or semi-independent function that measures risk, validates prices, monitors limits, benchmarks performance, and challenges front-office decisions.

### **Missing money**

Concern that market revenues may be insufficient to finance needed capacity because scarcity or investment signals are muted, risky, capped, or incomplete.

### **Model inventory**

Register of models, owners, purposes, inputs, outputs, limits, validation status, and retirement criteria.

### **Model register**

Inventory of models, owners, inputs, outputs, validation status, and decisions supported.

### **Model value at risk**

Scenario estimate of value exposed to a wrong model-driven action across affected MW, duration, adverse price error, and bad-event count.

### **Model-risk charge**

Scenario or governance allowance for losses, overrides, monitoring, and control burden introduced by relying on a model.

### **Multi-buyer PPA**

PPA where several buyers aggregate demand to contract with one generator or supplier structure.

### **Nameplate capacity**

Rated maximum power under defined conditions.

### **NEMO**

Nominated Electricity Market Operator: exchange/operator designated for day-ahead and intraday market coupling functions.

### **Netting**

Partial cancellation of positive and negative deviations inside compatible commercial or settlement boundaries.

### **NPV**

Net present value: discounted value of future cash flows minus investment cost.

### **Numerical weather prediction**

Physics-based weather model used as an input to power, load, price, and system forecasts.

### **Objective function**

Mathematical expression a model seeks to maximize or minimize.

### **Offtaker**

Buyer of electricity or environmental attributes under a commercial contract such as a PPA.

### **Opportunity cost**

Value of the best feasible alternative action forgone when an asset is committed to a different action.

### **Paid-or-required capability gap**

Difference between system-service capability that is required or valuable at a site and capability that is already qualified and available.

### **Partner benchmark**

Feasible counterfactual used to compare actual BRP/BSP/optimizer performance with actions available under real constraints and rights.

### **Pay-as-produced PPA**

PPA where the buyer purchases electricity as generated, so the buyer generally takes more generation-profile exposure.

### **Peak load**

Maximum load over a defined period and geography.

### **Perfect foresight dispatch**

Dispatch optimized with actual future information; useful as an upper bound, not a fair operational benchmark.

### **Physical withholding**

Making capacity unavailable even though it could physically produce.

### **Pivotal supplier**

Supplier whose capacity is needed to meet demand under the relevant constraints.

### **Portfolio exposure**

Sensitivity of portfolio value or cash flow to a market, physical, contractual, credit, or operational driver.

### **Portfolio PPA**

PPA backed by a portfolio of assets rather than a single identified plant, potentially spreading resource and operational risks.

### **Potential future exposure**

Forward-looking estimate of how large counterparty exposure may become under future market movements.

### **Power**

Rate of energy flow, usually measured in W, kW, MW, or GW.

### **PPA**

Power Purchase Agreement: contract for purchase/sale of electricity, often long-term.

### **Private-wire PPA**

PPA where generation and consumption are connected through a direct/private wire rather than only through ordinary grid supply arrangements.

### **Product-readiness score**

Heuristic assessment of whether a firm has the assets, portfolio, data, control rights, risk limits, and evidence needed to offer a more advanced PPA product.

### **Qualification chain**

Sequence that turns technical capability into commercial value: specify, test, qualify, control, measure, settle, and enforce.

### **Quote stack**

Pricing bridge that decomposes an offered PPA price into raw energy, shape/firming, imbalance, grid/tariff, certificates, credit, tenor, optionality, and risk-margin components.

### **Reactive power**

Component of AC power associated with voltage support and electric/magnetic fields, commonly measured in var/MVAr.

### **Reference price**

Market price, index, or formula used to settle a financial PPA, CfD, floor/cap, or indexed price component.

### **Regional upscaling**

Converting asset-level forecasts or samples into a portfolio, bidding-zone, or control-area forecast.

### **Reserve capacity**

MW held available for possible TSO activation under product rules.

### **Reserve headroom**

MW or MWh capability held back so a battery can deliver a reserve obligation if called.

### **Residual demand**

Demand left for a supplier after subtracting other available supply and feasible imports.

### **Residual load**

Load remaining after subtracting variable renewable generation or another specified supply category.

### **Residual obligation**

Hourly MWh, EUR, evidence, or operational exposure left after contracted clean supply, storage, hedges, certificates, and control actions have been applied.

### **Residual Supply Index**

Market-power screen comparing supply available without a supplier to demand; values below one indicate pivotality in the simple form.

### **Retained-risk table**

Decision-memo table naming risks that remain after the recommendation, their owners, warning signals, and mitigations.

### **Revenue quality ladder**

Course tool ranking modeled value by how confidently it can support fixed obligations versus equity upside or unproven optionality.

### **Revenue stack**

Set of revenue sources a flexible asset may pursue, subject to mutual exclusivity, opportunity cost, and operational constraints.

### **Risk limit**

Quantitative or qualitative boundary on positions, losses, exposures, stress losses, collateral needs, or permitted strategies.

### **Round-trip efficiency**

Energy discharged divided by energy charged over a cycle.

### **Scarcity duration**

Time length of the shortage or obligation a resource must bridge.

### **Scarcity pricing**

High pricing that reflects real scarcity of energy, capacity, flexibility, or network capability.

### **Scarcity-duration distribution**



Distribution showing how often scarcity or obligation events of different lengths occur, for example 1-hour, 4-hour, overnight, multi-day, or seasonal events.

### **Security of supply**

Broad ability of the power system to serve demand reliably across operating and planning horizons.

### **Shadow price**

Marginal value of relaxing a binding constraint by one unit in an optimization problem.

### **Shadow trading**

Counterfactual simulation of feasible trading/dispatch decisions used to benchmark actual execution or partner performance.

### **Shape cost**

Cost of converting asset output into a promised contract profile, including market purchases, storage losses, degradation, hedging, balancing, grid/tariff costs, and risk margin.

### **Shape risk**

Risk that production or consumption timing differs from the contracted or hedged delivery shape.

### **Short-circuit ratio**

Screening concept comparing grid strength to connected converter capacity; detailed definition depends on method.

### **State of charge**

Battery energy currently stored, usually expressed as MWh or percent of usable capacity.

### **Stress test**

Scenario analysis of severe but plausible or intentionally challenging market, credit, liquidity, or operational conditions.

### **Structural market power**

Market-power potential arising from market structure, location, ownership, concentration, constraints, or pivotality.

### **Subscribed capacity**

Capacity level contracted for grid use or tariff purposes.

### **System adequacy**

Longer-term ability of the system to have enough resources and network capability to meet demand under credible conditions.

### **System margin**

Difference between available dependable resources and demand or required reserve/adequacy target, depending on definition.

### **System security**

Short-term ability of the power system to remain balanced and withstand disturbances.

### **System-service option**

Capability that may create value through connection approval, compliance, avoided curtailment, paid services, or future grid requirements, but is not yet base-case revenue.

### **Terminal value**

Value assigned to the final state of a storage model, often to avoid artificial emptying at the end of the optimization horizon.

### **Termination amount**

Amount due on early contract termination, often linked to mark-to-market or alternative valuation methods.

### **Throughput budget**

Allowed or planned cumulative charged/discharged energy over a period or asset life.

### **Transmission**

High-voltage bulk power transport over the main grid.

### **TSO**

Transmission System Operator: entity responsible for secure operation of the transmission system and system balance within its mandate.

### **Use-of-network charge**

Recurring charge for grid use, potentially covering infrastructure, losses, system services, metering, or reactive charges depending on tariff design.

### **Value at Risk**

Loss threshold for a specified probability and horizon; it does not describe the severity of losses beyond the threshold.

### **Value of lost load**

Estimate of consumer willingness to pay to avoid involuntary interruption of electricity supply.

### **Value pool**

Source of possible economic improvement, such as better pricing, reduced imbalance, improved dispatch, or reduced leakage.

### **Value-duration curve**

Curve or table showing the marginal value of adding each extra hour of usable storage duration after efficiency, degradation, grid, tariff, contract, and opportunity-cost constraints.

### **Value-weighted error**

Forecast or model error weighted by the economic consequence of being wrong in that interval or scenario.

### **Voltage**

Electrical potential difference; grid voltage must remain within operational limits.

### **WACC**

Weighted average cost of capital: combined cost of debt and equity after weighting by capital structure and tax treatment where applicable.

### **Warranty envelope**

Operating conditions under which a battery's warranty, performance guarantee, availability commitment, or remedy structure remains valid.

### **Weather-dependent**

Output materially depends on weather conditions outside operator control.

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